Charotar University of Science and Technology
Chandubhai S Patel Institute of Technology
M & V Patel Department of Electrical Engineering

List of Experiments

Subject: Microprocessor & Microcontrollers (EE 246) (4^{th} Sem)

1. Introduction to 8085 and various registers. To understand and perform data transfer operations in 8085 and memory.
2. To understand and perform arithmetic operations using microprocessor 8085.
3. To understand and perform branching, looping operations using microprocessor 8085.
4. To Study C Programming concepts as follows.
    i. Declaration of Variable and Arrays: Single & Multi-Dimensional Arrays, Function declaration and call of function, Typecasting.
    ii. Various header files & Math Functions available in math.h, Microcontroller 8051 specific header file reg51.h & reg52.h.
    iii. Various Control Loops: if, if-else, Nested if-else, for, while, do-while, switch-case, go to.
    iv. Keyword available in Keil. (Especially sbit, sfr, interrupt), Assignment of SFR address via header file & sbit/sfr Approach.
    v. C Operators: Arithmetic & Logical Operators, Conditional Operators, Shift Operators
5. To use port as an Output Port of microcontroller 8051.
6. To use port as an Input Port of microcontroller 8051.
7. To generate time delay using timer0/1 in mode2.
8. To generate time delay using timer0/1 in mode1.
9. To use Timer module as counter.
10. To program 8051 for external interrupt.
11. To program serial communication of 8051.
12. Introduction to 16x2 LCD and programming for printing data on 16x2 LCD module.
13. Introduction to ADC0804/ADC0808 and programming for ADC0804/ADC0808.
14. Introduction to DAC0808 and programming for DAC0808.